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Presenter · slides 22–28 · about 6:16 of narration

> THE PROJECT STORY
>
> We wanted to study dispatch decisions. April's data described aggregate patterns rather than individual journeys, so we built plausible evenings from evidence and explicit assumptions. We checked the implementation, replayed identical evenings under different transfer rules, and found a conditional trade-off between timely service and Lyft use. Our next work tests how robust those findings are.

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1 Your section

Slides 22–28: The predictive rules reached more requests promptly—and used more Lyft → What we can defend today.

Purpose: Findings, transfer outcomes, Pareto trade-offs, leader–follower extension, limitations, next steps and conclusion.

You start immediately after the handoff sentence: "We then replayed the same evenings under different transfer rules. Vignesh will explain what changed, what it means and what remains uncertain." Begin with slide 22's first sentence; do not re-introduce yourself.

You end the talk with "Thank you; we welcome your questions."

2 Essential preparation reading

Read these in the shared guide (`01_Team_Presentation_Guide.pdf`, one folder up) and the methodology PDF:

- Slide 22 · The predictive rules reached more requests promptly—and used more Lyft: guide, slide 22 (guide p. 42); methodology §7.7 M3: the L5 pilot (mechanism 5: the 15-minute measure), p. 41
- Slide 23 · Moving a request to Lyft does not end its wait: guide, slide 23 (guide p. 44); methodology §7.7 M3: the L5 pilot (mechanism 1 and its interpretation), p. 41; §3.9 Lyft: what happens after a transfer, p. 18
- Slide 24 · Better service and lower Lyft use can point to different choices: guide, slide 24 (guide p. 46); methodology §7.8 Reading the results as trade-offs, p. 42
- Slide 25 · Could we choose the transfer policy systematically?: guide, slide 25 (guide p. 48); methodology §7.9 A leader–follower reading, p. 43
- Slide 26 · Which assumptions could change the answer?: guide, slide 26 (guide p. 50); methodology §7.11 Limitations to keep in view, p. 44; §2.3 Calibration keeps a set, not a point, p. 11
- Slide 27 · Test robustness before fine-tuning a winner: guide, slide 27 (guide p. 51); methodology §7.12 Next milestone and how the work grows, p. 45; §7.13 Open decisions for the team, p. 46
- Slide 28 · What we can defend today: guide, slide 28 (guide p. 52); methodology §7.10 Status by item, p. 44
- Everyone: guide sections “What every member must be able to explain” and “Quick Q&A navigation”.

3 Concepts you must explain confidently

- the main result and that the 15-minute measure favours early transfer
- why a transfer does not end waiting, and the corrected patience interpretation
- dominance, why vans only stays nondominated, and why nondominated is not recommended
- leader–follower as a proposed extension: what a finite search can and cannot show
- which assumptions could change the answer, the order of next experiments, and the conditional conclusion

4 Questions you lead on

Answer these first; route anything else to its owner (guide, Quick Q&A navigation).

- Results and the 15-minute measure — guide slides guide, slide 22 (guide p. 42); backups 35
- Losses after transfer; patience interpretation — guide slides guide, slide 23 (guide p. 44); backups 35
- Pareto dominance and its limits — guide slides guide, slide 24 (guide p. 46); backups 36
- Leader–follower / bilevel extension — guide slides guide, slide 25 (guide p. 48); backups 31
- Limits, next experiments, the conditional answer — guide slides guide, slide 26 (guide p. 50), guide, slide 27 (guide p. 51), guide, slide 28 (guide p. 52); backups 37

5 Your narration (exact words)

Slide 22 · The predictive rules reached more requests promptly—and used more Lyft

about 64 s · open slide

SAY (exact narration)

These are thirty constructed burst-demand nights with provisional rider behaviour. Each bar is the share of all requests a vehicle reached at the curb within fifteen minutes—curb arrival, not boarding or a completed trip. In this two-van pilot, the predictive rules reached about three quarters of requests within fifteen minutes, compared with about half under vans only. They also used roughly nine to ten and a half Lyft orders per night. The waiting-time rules added little, partly because a rider moved after ten minutes almost never counts within fifteen; on requests picked up, the order is the same. This comparison does not isolate the benefit of prediction at an equal Lyft budget.

TRANSITION

But reaching a rider with a Lyft is not the end of the story.

Slide 23 · Moving a request to Lyft does not end its wait

about 58 s · open slide

SAY (exact narration)

The rider still waits through ordering and Lyft arrival. Their original patience clock continues running. Under Wait ten minutes, 95 of 196 transferred requests cancelled before their Lyft arrived, some before it was even ordered; under the prediction rules, about a third. Riders moved by the prediction rules had usually waited less when they were moved. With our placeholder patience, where giving up becomes likelier the longer someone has waited, that is consistent with their lower share lost after transfer. It is a possible contributing mechanism, not an isolated cause, and whether the gap survives other patience assumptions is untested.

TRANSITION

So rules trade service against Lyft use. How do we compare them?

Slide 24 · Better service and lower Lyft use can point to different choices

about 66 s · open slide

SAY (exact narration)

A tested rule is dominated if another tested rule is at least as good on both selected objectives and strictly better on one. In the small picture, A dominates B; A and C do not dominate each other. On the thirty-night means, five of our six rules are not dominated. Vans only stays nondominated simply because it uses no Lyft. Unassigned for ten minutes differs from vans only by about one request in 840, so it is effectively a tie that uses slightly more Lyft; we apply no noise margin yet. Nondominated does not mean recommended. This is the nondominated set among six tested rules under these two objectives—not the complete frontier of possible policies. [optional] Adding objectives or criteria could change the assessment.

TRANSITION

Which raises a natural question.

Slide 25 · Could we choose the transfer policy systematically?

about 66 s · open slide

SAY (exact narration)

One possible extension is to optimize transfer policy while anticipating how the routing dispatcher responds. A policy designer chooses the rule and threshold; the dispatcher responds by re-planning; the outcomes inform the choice. This is proposed, not implemented. It would be bilevel because the dispatcher optimizes its own routing points, which differ from the designer's service and Lyft criteria. A finite search does find the best option in a finite set for the average over our sampled nights, but not over all policies or every continuous threshold. [optional] To state it precisely, we would still need to define the leader's choices, the follower's problem and timing, how ties are treated, the expectation over nights, and the weight on Lyft use.

TRANSITION

Before choosing anything, we need to know which assumptions matter.

Slide 26 · Which assumptions could change the answer?

about 54 s · open slide

SAY (exact narration)

The findings are conditional on the evenings and behaviours we constructed. Patience is the assumption we most expect to matter, though we have not tested that: it sets how many riders give up and the losses after transfer. Lyft delay: a slower or more variable Lyft could weaken transfers; under our placeholder patience, late transfers most. Workload: the pilot is deliberately heavy; at lighter load the rules may barely differ. Geography changes van reach and predictions. [optional] And the candidate restriction, three new pickups per van, shapes both van performance and the predictions the rules read.

TRANSITION

That ordering tells us what to test next.

Slide 27 · Test robustness before fine-tuning a winner

about 40 s · open slide

SAY (exact narration)

Next, we should test whether the findings survive different rider-patience and Lyft-delay assumptions, while checking feasibility at the larger fleet size. Then the team must agree how much extra Lyft use one more rider reached is worth, compute confidence intervals across the paired nights, and only then search over thresholds. Order matters: a threshold tuned to placeholder behaviour would be tuned to our assumptions, not to the service.

TRANSITION

So, what can we defend today?

Slide 28 · What we can defend today

about 28 s · open slide

SAY (exact narration)

Aggregate data did not give us individual journeys. It gave us constraints for building plausible evenings. Within our specified pilot, predictive transfer rules improved timely curb arrival while using more Lyft. Choosing a policy requires testing the assumptions and deciding what trade-off is acceptable. Thank you; we welcome your questions.

TRANSITION

(End of the talk; questions follow.)

6 Manual verification activity: One transfer timeline ("Lyft arrives sooner")

> **WHAT HUMAN VERIFICATION MEANS**
>
> A program can generate and check an answer. Human verification means a teammate independently works through a defined example, records their calculation and compares it with the implementation.

This is a guided preparation activity, not a quiz or an exam. Work through the procedure on your own first and write down your result. Only then turn to the reference answer, which is on a separate page. If your result differs, record the difference rather than adjusting your working to match: a disagreement is useful information. The reference answer was produced with AI assistance from the project files; it does not count as a teammate's verification.

Related slides: Slides 18 and 23; also slide 17. **Goal:** Work through the verified designed-case transfer independently: the expected Lyft arrival, the comparison with the predicted van pickup, the transfer margin, the rider's total wait, and why the van time is not an observed counterfactual.

Source inputs

Designed evening L4, rule "Lyft arrives sooner", request R011. On this small designed evening the Lyft delay is fixed at 8 minutes by design, so realized and expected Lyft times coincide.

Input	Value	Kind
Request time	18:16:30	Recorded event
Decision moment (rule evaluated)	18:17:00	Recorded event
Van plan's predicted pickup for R011 at 18:17:00	18:30:08	**Prediction**
Ordering lag (decision to Lyft order)	2 min	Assumed setting
Typical Lyft delay used by the rule	8 min	Assumed setting
Transfer margin	2 min	Rule parameter
Lyft ordered	18:19:00	Recorded event
Lyft at the curb	18:27:00	Recorded event
Rider dropped off by the Lyft	18:30:25	Recorded event

Sources (with repository access): `outputs/sim/L4_R3-2/epochs.jsonl` (decision at 1,020 s) and `outcomes.csv` (R011).

Procedure

1. Compute the expected Lyft pickup at the decision moment: decision time + ordering lag + typical delay.
2. Compute the difference between the predicted van pickup and the expected Lyft pickup, in minutes and seconds.

3. Decide whether the rule fires: is the Lyft expected *more than* 2 minutes sooner?
4. Compute the rider's wait from request to the vehicle's curb arrival.
5. Compare the drop-off time 18:30:25 with the predicted van pickup 18:30:08 and state the exact difference.
6. Explain in two sentences why the predicted van time cannot be used to claim that the transfer helped this rider.

Worksheet

YOUR WORKING

Step-by-step working (use the lines, or attach your own sheet)

My result

Record

Status	**Awaiting human completion**
Name	
Date completed	
My result	
Matches the reference? (yes / partly / no)	
Discrepancy, and what I did about it	
What remains unverified	

Reference answer

REFERENCE ANSWER (compare only after finishing your own working)

Expected Lyft pickup. 18:17:00 + 2 min + 8 min = **18:27:00**.

Difference. 18:30:08 − 18:27:00 = **3 min 8 s**. This is more than the 2-minute margin, so the rule fires and R011 is moved to Lyft. (Equivalently: the rule fires when the predicted van pickup is later than now + 12 minutes, that is, later than 18:29:00.)

Wait. The Lyft reached the curb at 18:27:00, so the wait was 18:27:00 − 18:16:30 = **10 min 30 s**.

Drop-off versus predicted van pickup. 18:30:25 is **17 seconds after** 18:30:08, not before it. A drop-off and a predicted pickup are different events, so this comparison shows nothing about which option was better.

Why the van time is not a counterfactual. 18:30:08 is the van plan's *prediction* at 18:17:00, made before later events. Once R011 moved to Lyft, the vans re-planned without it, so we never observe when, or whether, a van would actually have picked this rider up. The example shows how the rule compares two predictions; it is not evidence of a saving. Whole evenings are compared instead (slide 20).

Comparison checklist

- ☐ Did I include the 2-minute ordering lag in the expected Lyft time?
- ☐ Is my difference exactly 3 min 8 s, and did I compare it with the 2-minute margin using "more than"?
- ☐ Is my wait 10 min 30 s, measured to curb arrival, not to drop-off?
- ☐ Did I state that 18:30:25 is 17 seconds after 18:30:08?
- ☐ Did I explain that the van time is a prediction, not an observed outcome?

Record any disagreement in the table above. Do not change your own working to match the reference.

7 Rehearsal and readiness checklist

- ☐ Read my slides' preparation notes in the shared guide.
- ☐ Read my slides in the final deck (`02_Final_Presentation.pdf`) and know which backup slides I may need.
- ☐ Said my narration aloud with a stopwatch; my section took _____ (estimate 6:16).
- ☐ Practised my opening and handoff sentences with the neighbouring presenter.
- ☐ Can explain every concept listed above without the slides.
- ☐ Completed my verification activity and filled in the record (status changed from "Awaiting human completion" only by me).
- ☐ Took part in one timed full-team rehearsal (total time: _____).